LATTICE BASIS REDUCTION

0. OVERVIEW

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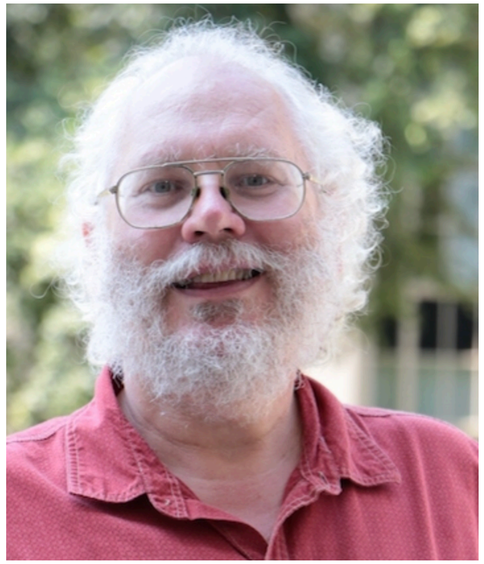
Public-key cryptographic schemes



- As of 2025, the public-key cryptographic schemes that are widely deployed in practice are RSA and elliptic curve cryptosystems (ECC).
- In 1994, Peter Shor discovered a very fast quantum algorithm that can break all RSA and elliptic curve systems.
- Although the timeline for building cryptographically-relevant quantum computers remains uncertain, interest in deploying quantum-safe cryptosystems has surged in the last two years.

Kyber and Dilithium

- In August 2024, the U.S. government's National Institute of Standards and Technology (NIST), published standards for quantum-safe lattice-based schemes:
  - Kyber (ML-KEM): A key encapsulation mechanism (KEM).
  - Dilithium (ML-DSA): A digital signature scheme.
- Kyber and Dilithium are already being deployed in large-scale applications, and the rate of deployment will substantially increase in the next couple of years.

Security of Kyber and Dilithium

- The security of Kyber and Dilithium is based on the hardness of solving two problems:
  - Learning With Errors (LWE) problem.
  - Shortest Integer Solutions (SIS) problem.
- Both LWE and SIS can be reformulated as instances of the Shortest Vector Problem (SVP) and its approximate version.
  - SVP is the problem of finding a nonzero vector of shortest length in a lattice.
  - The fastest methods known for solving SVP are refinements of the Lenstra-Lenstra-Lovász (LLL) algorithm.

Lattice basis reduction

- LLL was discovered in 1982 by Arjen Lenstra, Hendrik Lenstra, and László Lovász.
- It's a fast algorithm for finding a relatively short basis for a lattice.
- The LLL algorithm (and its refinements, including the BKZ algorithm) has many applications in cryptography.
  - Most importantly, it's used to estimate the security level of lattice-based cryptographic schemes such as Kyber and Dilithium.
- In much the same way as a solid grasp of the Number Field Sieve and Pollard's rho algorithm is essential for understanding the security of RSA and ECC, *a solid understanding of the LLL algorithm is key to appreciating the security foundations of Kyber and Dilithium.*

Course outline

This course will give a detailed explanation of the LLL lattice basis reduction algorithm.

1. Introduction to lattices
2. Gauss reduction
3. Gram-Schmidt orthogonalization
4. The LLL algorithm
5. Cryptographic applications
6. LLL improvements

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Related courses on post-quantum cryptography

1. **Kyber and Dilithium**
   - Detailed descriptions of how Kyber and Dilithium work.
2. **The Mathematics of Lattice-Based Cryptography**
   - Introduces the SIS and LWE problems, and explains why they are considered to be lattice problems. Also shows their connection to SVP.
3. **Hash-Based Signatures**
   - Introduction to quantum-safe signature schemes based on hash functions, including LMS and SPHINCS+.

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